

The Silent Tax: Typo-Robustness Analysis (ARC)

DeepSeek-R1-Distill-Qwen-7B

All tables are *answered-only*: traces that hit the token cap before writing a final answer are excluded from the metrics. Configs are named **typo{rate}_real{ratio}**: *rate* = % of words corrupted, *real* = % of typos that form real words. Baseline is the clean input.

Accuracy and Flips

Table 1: Accuracy per config: answered-only and strict (truncated=wrong), with 95% bootstrap CI on acc_strict.

config	n	acc_answered	acc_strict	ci_lo	ci_hi
clean	500	0.8505	0.774	0.738	0.81
typo25_real10	500	0.8645	0.842	0.808	0.874
typo25_real40	500	0.856	0.832	0.798	0.862
typo25_real70	500	0.82	0.802	0.764	0.836
typo50_real10	500	0.8193	0.798	0.762	0.832
typo50_real40	500	0.8308	0.766	0.728	0.802
typo50_real70	500	0.8244	0.798	0.76	0.834
typo75_real10	500	0.8074	0.788	0.754	0.822
typo75_real40	500	0.7732	0.75	0.71	0.788
typo75_real70	500	0.7469	0.726	0.686	0.766

Table 2: Answered-only flips vs clean, paired by question over n_both (answered in both conditions). r2w / w2r = questions correct on clean but wrong on this config / wrong on clean but correct on this config; r2w_ratio = r2w / n_both, w2r_ratio = w2r / n_both; net_flip_ratio = r2w_ratio - w2r_ratio = (r2w - w2r) / n_both, the net rate at which this config worsens answers relative to clean.

config	n_both	r2w	w2r	r2w_ratio	w2r_ratio	net_flip_ratio	mcnemar_b	mcnemar_c	mcnemar_stat	mcnemar_p
typo25_real10	443	21	27	0.0474	0.0609	-0.0135	21	27	0.521	0.47048642
typo25_real40	442	25	31	0.0566	0.0701	-0.0136	25	31	0.446	0.50403587
typo25_real70	444	39	25	0.0878	0.0563	0.0315	39	25	2.641	0.10416256
typo50_real10	442	38	26	0.086	0.0588	0.0271	38	26	1.891	0.16913144
typo50_real40	429	37	25	0.0862	0.0583	0.028	37	25	1.952	0.16241315
typo50_real70	441	42	27	0.0952	0.0612	0.034	42	27	2.841	0.09191105
typo75_real10	446	52	32	0.1166	0.0717	0.0448	52	32	4.298	0.03816577
typo75_real40	441	59	26	0.1338	0.059	0.0748	59	26	12.047	0.00051874
typo75_real70	444	73	23	0.1644	0.0518	0.1126	73	23	25.01	5.7e-07

McNemar test (used in the flips table above).

- **What it is:** a paired significance test for before/after binary outcomes. It looks only at the questions that *changed* (the discordant pairs: *b* = right→wrong, *c* = wrong→right) and ignores those that stayed the same.

- **Null hypothesis:** typos break correct answers as often as they fix wrong ones ($b = c$, i.e. no net effect).
- **Statistic:** $(|b - c| - 1)^2 / (b + c)$ (column `mcnemar_stat`); `mcnemar_p` is its chi-square p -value with 1 degree of freedom.
- **When we reject it:** when `mcnemar_p` < 0.05 — the right→wrong vs wrong→right imbalance is too large to be chance.

Reasoning Length

Table 3: Absolute generated tokens per config (answered-only).

config	n_answered	mean_tok	median_tok	p90_tok	max_tok
clean	455	1153.9	824	2235	8512
typo25_real10	487	1221.9	869	2400	8753
typo25_real40	486	1245.6	872.5	2588	8684
typo25_real70	489	1233.8	871	2680	7092
typo50_real10	487	1298.2	936	2615	9068
typo50_real40	461	1320.7	868	2814	9110
typo50_real70	484	1329.2	922.0	2752	8588
typo75_real10	488	1380.7	963.0	2752	7480
typo75_real40	485	1412.8	930	3084	7686
typo75_real70	486	1452.9	998.0	3154	11834

Table 4: Mean generated tokens for correct vs wrong answers. `wrong_over_correct` = `tok_wrong` / `tok_correct` (mean tokens of wrong answers divided by mean tokens of correct answers).

config	tok_correct	tok_wrong	wrong_over_correct
clean	999.9	2030.3	2.0306
typo25_real10	1120.3	1870.4	1.6695
typo25_real40	1114.3	2025.7	1.8179
typo25_real70	1066.2	1997.5	1.8735
typo50_real10	1184.4	1814.5	1.532
typo50_real40	1174.5	2038.9	1.736
typo50_real70	1179.6	2031.6	1.7222
typo75_real10	1231.6	2005.7	1.6286
typo75_real40	1262.6	1924.8	1.5244
typo75_real70	1262.6	2014.7	1.5957

Self-doubt

The self-doubt markers are curated words grouped into two *banks*, each matched as a case-insensitive whole-word pattern in the reasoning text. Density is matches per 1000 reasoning words.

- **second_guess** — the proposal’s canonical self-correction words plus direct reversals: *wait, hmm, actually, reconsider, but wait, hold on, on second thought, let me recheck, double-check.*
- **uncertainty** — hedging / “not certain” words: *maybe, perhaps, possibly, might be, could be, not sure, I guess, presumably, or maybe / alternatively, unclear / confused.*

Example reasoning: “*Wait, maybe the total is 12... hmm, actually let me recheck*” contributes *wait*, *hmm*, *actually* (second_guess) and *maybe* (uncertainty). A separate *typo-noticing* bank (*typo*, *misspelled*, *doesn't make sense*, ...) is treated as repair behaviour, not self-doubt.

Table 5: Doubt-marker density per config: second_guess and uncertainty, per 1k words and per trace.

config	n_answered	second_guess_per_1k	second_guess_per_trace	uncertainty_per_1k	uncertainty_per_trace	total_per_1k	total_per_trace
clean	455	11.087	8.798	13.599	10.791	24.686	19.589
typo25_real10	487	11.807	9.916	14.511	12.187	26.318	22.103
typo25_real40	486	11.012	9.377	13.399	11.409	24.411	20.786
typo25_real70	489	10.85	9.245	14.574	12.419	25.424	21.665
typo50_real10	487	10.975	9.735	14.69	13.031	25.665	22.766
typo50_real40	461	11.611	10.562	15.886	14.451	27.497	25.013
typo50_real70	484	11.101	10.225	16.141	14.868	27.242	25.093
typo75_real10	488	11.545	10.969	17.063	16.213	28.608	27.182
typo75_real40	485	11.337	11.027	18.354	17.852	29.692	28.878
typo75_real70	486	11.642	11.675	19.197	19.251	30.839	30.926

Table 6: Per-marker density (markers per 1000 reasoning words): clean vs pooled-typo. discrimination = typo minus clean (larger = more typo-responsive). All bank markers, ranked by discrimination.

marker	category	clean_per_1k	typo_per_1k	discrimination
maybe	uncertainty	6.7	7.466	0.766
or maybe/alt	uncertainty	1.85	2.608	0.758
perhaps	uncertainty	1.415	1.895	0.48
wait	second_guess	5.905	6.25	0.345
unclear/confused	uncertainty	0.601	0.804	0.203
might be	uncertainty	0.85	0.923	0.073
could be	uncertainty	0.487	0.549	0.061
hold on	second_guess	0.199	0.257	0.057
not entirely	uncertainty	0.266	0.318	0.052
not sure	uncertainty	1.335	1.386	0.051
I guess	uncertainty	0.044	0.062	0.018
I assume	uncertainty	0.006	0.022	0.017
possibly	uncertainty	0.036	0.044	0.008
let me recheck	second_guess	0.169	0.175	0.006
on second thought	second_guess	0.0	0.0	0.0
presumably	uncertainty	0.008	0.006	-0.003
reconsider	second_guess	0.017	0.01	-0.007
but wait	second_guess	0.947	0.932	-0.016
double-check	second_guess	0.291	0.268	-0.022
wait/actually no	second_guess	0.496	0.471	-0.025
hmm	second_guess	2.304	2.261	-0.043
actually	second_guess	0.759	0.701	-0.058

Table 7: Doubt density (markers per 1000 reasoning words) split by final outcome, per category (2g = second.guess, un = uncertainty, tot = both). For a group, density = total markers in that group / total reasoning words in that group x 1000. *_correct = over correct-answer traces; *_wrong = over wrong-answer traces; *_wrong_over_correct = _wrong / _correct.

config	2g_correct	2g_wrong	2g_wrong_over_correct	un_correct	un_wrong	un_wrong_over_correct	tot_correct	tot_wrong	tot_wrong_over_correct
clean	10.427	12.794	1.227	11.322	19.483	1.721	21.748	32.277	1.484
typo25_real10	11.063	14.54	1.314	12.939	20.29	1.568	24.002	34.83	1.451
typo25_real40	10.774	11.76	1.092	11.436	19.581	1.712	22.21	31.341	1.411
typo25_real70	10.276	12.145	1.182	11.322	21.921	1.936	21.598	34.066	1.577
typo50_real10	10.716	11.753	1.097	13.051	19.626	1.504	23.767	31.379	1.32
typo50_real40	11.373	12.254	1.077	12.817	24.172	1.886	24.19	36.426	1.506
typo50_real70	10.664	12.242	1.148	13.617	22.733	1.669	24.281	34.975	1.44
typo75_real10	11.54	11.557	1.002	14.991	22.133	1.476	26.531	33.69	1.27
typo75_real40	11.352	11.307	0.996	16.328	22.601	1.384	27.679	33.908	1.225
typo75_real70	11.637	11.651	1.001	16.419	24.121	1.469	28.056	35.772	1.275

LLM judge. The proposal specifies a second, non-lexical measure for both dimensions: a separate model with a fixed, strict prompt. The judge is a different model from the one under test, called at temperature 0 on a seeded random subsample of answered traces per config. It is shown the corrupted-word list (*original* $\rightarrow$ *as shown*), the question, and the reasoning trace (head+tail window), and must return JSON with one repair label, a 0–4 self-doubt rating, and a *verbatim quote* from the trace as evidence for each; rows whose quote is not found in the trace are flagged. Repair labels: *silent_readthrough* (never mentions the corruption), *explicit_notice_fix* (flags it and recovers the intended word), *explicit_notice_nofix* (flags it but never resolves it), *misread_wrong_word* (reasons from a different real word), *derailed* (stuck on the corruption), *unclear*, *not_applicable* (clean). Doubt rating: 0 = no hesitation, 2 = re-checks a step, 4 = pervasive doubt / cannot settle. The judge is a subsample measure; the lexical tables above remain the full-data result, and the agreement tables say how far the two independent measures coincide.

Table 8: LLM-judge self-doubt rating per config on the judged subsample. mean_rating = mean 0-4 rating; frac_ge2 / frac_ge3 = share of traces rated at least 2 / at least 3; marker_per_1k = the lexical doubt density of the SAME traces, for side-by-side comparison.

config	n	mean_rating	frac_ge2	frac_ge3	marker_per_1k
clean	324	2.494	0.9383	0.5679	19.724
typo25_real10	373	2.592	0.9598	0.6408	21.314
typo25_real40	369	2.621	0.9837	0.6369	20.665
typo25_real70	376	2.59	0.9601	0.633	21.6
typo50_real10	389	2.671	0.9743	0.7018	21.625
typo50_real40	347	2.571	0.951	0.6282	21.786
typo50_real70	369	2.645	0.9648	0.6775	22.956
typo75_real10	379	2.678	0.9842	0.6939	23.515
typo75_real40	401	2.671	0.9776	0.6783	23.518
typo75_real70	413	2.68	0.9709	0.7094	25.128

Table 9: LLM-judge doubt rating split by final outcome. $\text{rating_correct} / \text{rating_wrong} = \text{mean rating over correct} / \text{wrong traces}$; $\text{wrong_over_correct} = \text{rating_wrong} / \text{rating_correct}$.

config	rating_correct	rating_wrong	wrong_over_correct	n_correct	n_wrong
clean	2.435	2.782	1.142	269	55
typo25_real10	2.557	2.8	1.095	318	55
typo25_real40	2.573	2.867	1.114	309	60
typo25_real70	2.532	2.827	1.117	301	75
typo50_real10	2.639	2.803	1.062	313	76
typo50_real40	2.513	2.809	1.118	279	68
typo50_real70	2.592	2.853	1.101	294	75
typo75_real10	2.635	2.831	1.074	296	83
typo75_real40	2.627	2.817	1.073	308	93
typo75_real70	2.605	2.897	1.112	306	107

Table 10: Convergent validity of the two self-doubt measures: per-trace correlation between the judge’s 0-4 rating and the lexical marker density, per config and pooled (`_all_`). Positive values mean the independent measures agree.

config	n	pearson	spearman
clean	324	0.5369	0.6082
typo25_real10	373	0.5274	0.6251
typo25_real40	369	0.4945	0.5271
typo25_real70	376	0.5554	0.6339
typo50_real10	389	0.4811	0.5632
typo50_real40	347	0.547	0.6186
typo50_real70	369	0.561	0.6213
typo75_real10	379	0.5353	0.5983
typo75_real40	401	0.5296	0.5994
typo75_real70	413	0.5269	0.5974
<code>_all_</code>	3740	0.5286	0.6019

Repair Behaviour

Table 11: How each corrupted word was handled, per config (categories defined below the table).

config	n_corrupt_words	silent_fix_n	flagged_n	misread_n	not_used_n	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
typo25_real10	1964	1572	114	41	237	0.8004	0.058	0.0209	0.1207
typo25_real40	1981	1481	180	61	259	0.7476	0.0909	0.0308	0.1307
typo25_real70	1988	1384	243	105	256	0.6962	0.1222	0.0528	0.1288
typo50_real10	3932	3023	317	100	492	0.7688	0.0806	0.0254	0.1251
typo50_real40	3747	2691	439	135	482	0.7182	0.1172	0.036	0.1286
typo50_real70	3887	2594	551	208	534	0.6674	0.1418	0.0535	0.1374
typo75_real10	5847	4154	784	196	713	0.7104	0.1341	0.0335	0.1219
typo75_real40	5777	3802	944	291	740	0.6581	0.1634	0.0504	0.1281
typo75_real70	5769	3429	1174	398	768	0.5944	0.2035	0.069	0.1331

A *corrupted word* is a word that received a typo — it differs between the clean and typo question (e.g. *sum* $\rightarrow$ *sun*). We recover these by diffing the two questions, then place each into one bucket by which form appears in the reasoning:

- **silent_fix** — only the original word (*sum*) appears: a quiet recovery.
- **flagged** — both forms appear (*sum* and *sun*): noticed and fixed.

- **misread** — only the corrupted form (*sun*) appears: read as the wrong word.
- **not_used** — neither form appears in the reasoning.

Columns ***_n** are counts; ***_frac** are the share of **n_corrupt_words**.

Table 12: Word handling pooled by real-word ratio: the non-word vs real-word contrast.

real_ratio	n_corrupt_words	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
10	11743	0.745	0.1035	0.0287	0.1228
40	11505	0.6931	0.1359	0.0423	0.1287
70	11644	0.6361	0.169	0.0611	0.1338

Table 13: Misread rate split by final correctness. A typo’d word is ‘misread’ when the reasoning uses only its corrupted form (e.g. *sun* instead of *sum*). Within each outcome group, misread rate = (typo’d words that were misread) / (all typo’d words examined), pooled over the group. misread_correct = over problems answered correctly; misread_wrong = over problems answered wrong; wrong_over_correct = misread_wrong / misread_correct.

config	misread_correct	misread_wrong	wrong_over_correct
typo25_real10	0.0178	0.0391	2.196
typo25_real40	0.0269	0.0521	1.939
typo25_real70	0.0501	0.0641	1.28
typo50_real10	0.0201	0.0485	2.418
typo50_real40	0.033	0.0496	1.502
typo50_real70	0.0515	0.0615	1.193
typo75_real10	0.0293	0.0516	1.76
typo75_real40	0.0466	0.062	1.328
typo75_real70	0.0606	0.0931	1.535

Table 14: LLM-judge repair category per config (share of judged traces per label; labels defined in the note in the Self-doubt section). Unlike the word-level table above, the unit here is the whole trace, not a single corrupted word.

config	n	silent_readthrough	explicit_notice_fix	explicit_notice_nofix	misread_wrong_word	derailed	unclear	not_applicable
clean	324	0.0	0.0031	0.0031	0.0	0.0	0.0	0.9938
typo25_real10	373	0.7587	0.1716	0.0161	0.0536	0.0	0.0	0.0
typo25_real40	369	0.7425	0.1653	0.0325	0.0596	0.0	0.0	0.0
typo25_real70	376	0.6755	0.1862	0.0293	0.109	0.0	0.0	0.0
typo50_real10	389	0.7326	0.2134	0.0411	0.0129	0.0	0.0	0.0
typo50_real40	347	0.6484	0.2709	0.0317	0.049	0.0	0.0	0.0
typo50_real70	369	0.5772	0.2683	0.0488	0.103	0.0	0.0027	0.0
typo75_real10	379	0.5752	0.3694	0.0343	0.0211	0.0	0.0	0.0
typo75_real40	401	0.5511	0.3566	0.0324	0.0574	0.0025	0.0	0.0
typo75_real70	413	0.4504	0.4165	0.0654	0.0678	0.0	0.0	0.0

Table 15: LLM-judge repair category pooled by real-word ratio, rates pooled. The proposal predicts misread_wrong_word rises and the explicit_notice_* labels fall as the real-word share grows.

real_ratio	n	silent_readthrough	explicit_notice_fix	explicit_notice_nofix	misread_wrong_word	derailed	unclear	not_applicable
10	1141	0.6889	0.2515	0.0307	0.0289	0.0	0.0	0.0
40	1117	0.6446	0.2668	0.0322	0.0555	0.0009	0.0	0.0
70	1158	0.5639	0.2945	0.0484	0.0924	0.0	0.0009	0.0

Table 16: Accuracy within each judge label over typo configs. accuracy = share of traces with that label that answered correctly; share = that label’s share of all judged typo traces. Low accuracy on silent_readthrough / misread_wrong_word is the confident-silent-failure case.

label	n	accuracy	share
silent_readthrough	2159	0.8411	0.632
explicit_notice_fix	926	0.7721	0.2711
explicit_notice_nofix	127	0.5906	0.0372
misread_wrong_word	202	0.5792	0.0591
derailed	1	1.0	0.0003
unclear	1	0.0	0.0003

Table 17: Agreement between the two independent repair measures on the same traces, after collapsing the judge labels onto the word-level buckets (silent_readthrough -> silent_fix, explicit_notice_* -> flagged, misread_wrong_word -> misread). The first rows are the confusion matrix (rows = judge, columns = word-level); the __summary__ row gives n_comparable, raw_agreement, cohen_kappa, and evidence_verbatim_frac (share of judged traces whose quoted evidence was found verbatim in the trace).

judge	wordlevel_silent_fix	wordlevel_flagged	wordlevel_misread	n_comparable	raw_agreement	cohen_kappa	evidence_verbatim_frac
silent_fix	1311	517	326				
flagged	181	425	446				
misread	42	73	86				
__summary__				3407	0.5348	0.2355	0.8567

Real-word Effect

This section isolates the effect of typo *kind* (real-word vs non-word), holding the question, corrupted positions, and rate fixed — something the vs-clean accuracy section cannot do. Three views:

- a **controlled paired** comparison of low-real vs high-real variants on the *same* questions, with a McNemar test (paired table);
- a **per-typo logistic regression** comparing the marginal harm of one real-word vs one non-word typo (logit table);
- whether wrong answers **flag the corruption** more or less as the real-word share rises (silent-failure table).

Read the signs, deltas and *p*-values in the tables below for this dataset’s result.

Table 18: Controlled paired comparison of two TYPO variants against EACH OTHER (baseline is the low-real variant, NOT clean – so this is not the vs-clean accuracy table). Within a fixed rate the same questions are corrupted in the same positions, only the typo KIND differs. Restricted to questions answered in both variants. $\text{acc_low_real} / \text{acc_high_real}$ = accuracy on the lower / higher real-word variant; $\text{delta_acc} = \text{acc_high_real} - \text{acc_low_real}$; r2w = questions correct in the low-real variant but wrong in the high-real one; w2r = the reverse; mcnemar_p = McNemar test on that paired change. Negative delta with $\text{r2w} \neq \text{w2r}$ means real-word typos are more harmful than the non-word typos they replaced.

rate	compare	n_both	acc_low_real	acc_high_real	delta_acc	r2w	w2r	mcnemar_p
25	real10_to_real70	476	0.8655	0.8277	-0.0378	46	28	0.04813101
25	real10_to_real40	474	0.865	0.8608	-0.0042	32	30	0.8989403
25	real40_to_real70	477	0.8616	0.8239	-0.0377	40	22	0.03085
50	real10_to_real70	472	0.8263	0.8326	0.0064	38	41	0.8219656
50	real10_to_real40	450	0.8311	0.8333	0.0022	33	34	1.0
50	real40_to_real70	448	0.8438	0.8281	-0.0156	39	32	0.47642205
75	real10_to_real70	475	0.8126	0.7453	-0.0674	60	28	0.00095108
75	real10_to_real40	474	0.8143	0.7743	-0.0401	54	35	0.05639171
75	real40_to_real70	471	0.7834	0.7495	-0.034	60	44	0.141326

Table 19: Logistic regression $\text{correct} \sim \text{num_real} + \text{num_nonword}$ over all answered typo traces ($\text{num_real} / \text{num_nonword}$ = count of real-word / non-word typos in the problem). coef_* = change in log-odds of a correct answer per one added typo of that kind (more negative = more harmful); $\text{odds_*} = \exp(\text{coef})$; intercept is the model constant. n = number of traces fit.

n	coef_num_real	coef_num_nonword	odds_real	odds_nonword	intercept
4353	-0.04173	-0.00804	0.9591	0.992	1.67484

Table 20: Among WRONG answers only, does the model flag the corruption, by real ratio? $\text{explicit_notice_frac}$ = fraction of wrong-answer traces whose reasoning uses strong typo-flagging language (typo / misspelled / doesn't make sense / ...); $\text{silent_frac} = 1 - \text{explicit_notice_frac}$. The proposal predicts real-word failures are quieter ($\text{explicit_notice_frac}$ should fall as real rises).

real_ratio	n_wrong	explicit_notice_frac	silent_frac
10	248	0.3548	0.6452
40	258	0.376	0.624
70	296	0.3986	0.6014

Lexical Grid (overview)

Table 21: All four marker families (second_guess, uncertainty, typo_noticing, repair_words) across configs.

config	second_guess	uncertainty	typo_noticing	repair_words
clean	11.087	13.599	1.036	0.748
typo25_real10	11.807	14.511	1.416	0.922
typo25_real40	11.012	13.399	1.51	1.037
typo25_real70	10.85	14.574	1.555	1.056
typo50_real10	10.975	14.69	1.588	1.012
typo50_real40	11.611	15.886	1.705	1.104
typo50_real70	11.101	16.141	1.736	1.099
typo75_real10	11.545	17.063	1.926	1.199
typo75_real40	11.337	18.354	2.181	1.376
typo75_real70	11.642	19.197	2.407	1.524

Spellcheck Recovery